The Australian Cardiovascular disease Data Commons (ACDC)

Data Sharing Policy

Table of Content

1. Background 2
2. Purpose 2
3. Scope 2
4. Definitions 3
5. Data access 4
   5.1 Data Access Procedure 4
      5.1.1 Data Access Workflow 5
6. Data Access Applications 6
   6.1 Eligibility 6
   6.2 Review and authorisation 7
   6.3 Project Schedules 7
7. Terms of use 8
   7.1 Data Access 8
   7.2 Terms 8
8. Data Access procedures 9
9. Dispute resolution 9
10. Data usage auditing 9

Additional Resources 9

References 10

Revision History 11

1. Background

The Australian Cardiovascular disease Data Commons (ACDC) platform has been established by leading Australian cardiovascular researchers to bring together well-characterised phenotyped cohort studies from across Australia. The platform will facilitate discoverability, access, sharing and reuse of chronic disease health and -omics datasets to support the research community in the endeavour to reduce the burden of cardiovascular disease and improve outcomes for patients.

The ACDC is an initiative of the Australian Cardiovascular Alliance (ACvA) Precision Medicine Flagship and the Australian BioCommons, and is administered by the Baker Heart and Diabetes Institute. The ACvA Precision Medicine Flagship is a group of collaborating research partners. The ACDC will facilitate a translational pipeline, to ensure the deployment and evaluation of these predictors within clinical practice. Significantly, this will be Australia's first global standards-based human data commons and will yield infrastructure that will directly inform and other data intensive precision medicine programs. other data intensive precision medicine programs.

2. Purpose

The purpose of this Policy is to describe the terms and processes required to facilitate appropriate, legal, and responsible sharing of data held within the ACDC.

The ACDC is committed to facilitating the responsible sharing of research data, in a manner that:

- Recognises the ownership of the datasets by the relevant Data Custodians (study steering committees or other bodies).
- Increases the benefits and reduces the risks associated with clinical research data sharing [1].
- Respects the contributions, consent, and privacy of individual research participants whose data are shared [1].
- Supports the FAIR guiding principles to ensure research data is Findable, Accessible, Interoperable and Reusable [2].
- Uses the Five Safes Framework to guide the assessment and management of risks associated with sharing and release of data [3].
- Complies with relevant ethical, regulatory and legal requirements such as the Australian Code for the Responsible Conduct of Research 2018 [4] and National Statement on Ethical Conduct in Human Research, 2007 (updated 2018). [5].
- Acknowledges Australian and international initiatives to improve access to and re-use of publicly-funded research data collections, including the OECD Principles and Guidelines for Access to Research Data from Public Funding 2007 and the Joint Statement on Sharing Research Data to Improve Public Health (Wellcome Trust).

3. Scope

This Data Sharing Policy pertains to the ACDC. This policy is applicable to the sharing of demographic, clinical, imaging and/or omics data in the form of aggregated and/or individual, de-identified participant data.

The ACDC includes a variety of demographic, clinical and omics data from a number of studies. Over time additional studies and data types may also be included in the ACDC. No identifiable data will be shared at any time within the ACDC.

4. Definitions

ACDC Platform Governance Committee:	Representatives nominated by each ACDC Consortium Member to provide leadership for and supervision of the ACDC and its investigators. The ACDC Platform Governance Committee may also include representatives from each Data Custodian.
ACDC Data Access Committee:	A committee of stakeholders, including researchers and representatives from data custodians, responsible for processing data requests and ensuring adherence of Users to this Policy.
ACDC Investigator Team:	The group of researchers who have made a substantial and ongoing contribution to the development and implementation of the ACDC including the Data Custodians and other parties responsible for the research from which data is requested.
Data Set	Data set that has been deposited by a Data Custodian with the ACDC using a Data Deposit Agreement.
Data Custodians:	Representative from the committee that has oversight of a particular study data set and who is employed by the organisation that is responsible for ongoing management of that particular study data set. The Data Custodian i responsible for providing the Data Set to ACDC and authorising data access to Data Applicants.
Data Sharing Agreement:	General term for an agreement between parties for the transfer of data from one party to another. This includes a Data Access Agreement and a Data Deposit Agreement – in all cases to be signed by an authorised delegate from each institution: the data recipient and the data provider – that specifies the terms of use of the requested data.
IPD:	Individual Participant-level Data.
Data Applicant	The User responsible for managing the proposed research project that is the subject of the Data Access Application and whose employing organisation will sign the Data Access Agreement.
Users	Research personnel employed by an organisation that register and are approved to use the ACDC and who have accepted the ACDC terms and conditions.

Data Access Application	Application form that is accessible via the ACDC platform wherein a User as a Data Applicant will provide the requested information about a proposed use of data to be accessed by way of the ACDC platform.
Data Access Agreement	Agreement between the organisation employing a Data Applicant and the Baker Heart and Diabetes Institute, acting on behalf of the Data Custodian, to provide legal terms and conditions for access to a requested data set.
Project Schedule	Agreement between the organisation employing a Data Applicant and the Baker Heart and Diabetes Institute, acting on behalf of the ACDC, to provide legal terms and conditions for access to multiple requested data sets within a single project.
Framework Agreement	Agreement between all Members of the ACDC Consortium to provide legal basis for data sharing and collaboration. A Data Custodian must be a Member of the ACDC Framework Agreement before the Custodian's data set can be accessed through the ACDC.

5. Data access

Curated data held within the ACDC is findable and searchable through the ACDC web interface which is currently under development. Data access requests are facilitated by the system. Features include access controls (such that appropriate clinical, demographic data and omics data is only available to authorised individuals) and search interfaces. The ACDC stores a core set of demographic and clinical data as well as an index of pointers to where omics data files are stored on a secure cloud storage. The ACDC also includes an associated secure analysis environment (for the analysis of clinical and demographic data). The ACDC and its analysis environment are hosted on secure cloud computing resources.

Access to data is managed through a tiered series of approval processes to ensure good research governance and to manage risks of non-adherence to this policy.

All Users will be required to agree to a set of terms and conditions that will apply to use of the ACDC platform. This agreement will be executed at or before User Registration and is necessary before any form of data access will be permitted.

5.1 Data Access Procedure

1st tier of access- summary information
Investigators and researcher collaborators can freely explore summary information about each study via the "Discovery" tab and view the structure and variety of fields collected in each study via the "Data Dictionary Explorer."

2nd tier of access - aggregated data

To explore the data held in the ACDC in more detail investigators and research collaborators must first complete a registration form that records their identity and contact details.

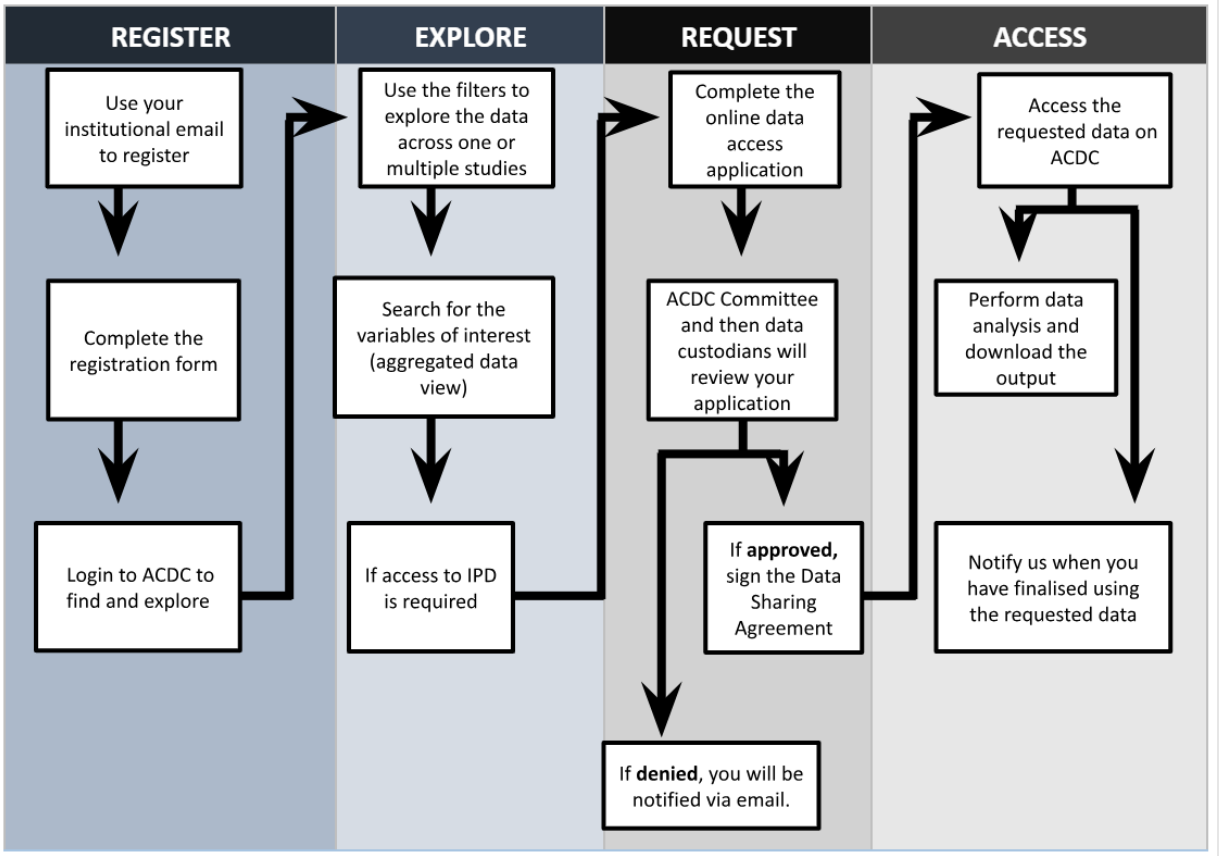
- Investigators and research collaborators must register using an email address from their organisation to become ACDC Users.
- Users can explore the ACDC using demographic and clinical filters for the searchable variables listed in the Appendix, which can be applied to refine queries. However, results will not be provided for queries that yield data on fewer than 50 individuals.
- Users can search for the variables across one or multiple studies and view aggregated data.

3rd Tier of access - individual participant data (IPD)

- Users can request access to individual participant data (IPD) from one or more data cohorts by submitting an online Data Access Application. This form prompts users to provide details required by the ACDC Data Governance Committee and the Data Custodians to access the application for compliance with policies, Agreements and Australian law. The form also details what data variables the User is requesting to access.

- The Data Access Application is reviewed by the Data Access Committee for compliance with this Policy. The outcome of this review and the application will be distributed to the Data Custodians for consideration.
- The Data Access Application is reviewed by the Data Custodian.
- If validated by the ACDC Data Access Committee and approved by the Data Custodians, the Users will be required to enter a Data Access Agreement / Project Schedule with the ACDC. The final Data Access Agreement is provided to the ACDC Governance Committee and User access is implemented.
- Users are allowed to access the specific data types (e.g. demographic and clinical variables, whole exome sequence, genome wide genotyping, metabolomics etc.) as requested in their data access application and approved by each trial's Data Custodian.
- Users can perform data analysis within the secure analysis environment and download the output of the analysis. Data will be stored within and securely downloaded from a secure cloud storage that is operated by the Australian BioCommons.
- If data is required to be downloaded into the User's own computing environment for analysis, this must be agreed to by the Data Custodian, or authorised delegate, and documented under the Data Sharing Agreement. Recipients of downloaded data must adhere to data return, retention, and destruction clauses in the Data Sharing Agreement.
- In some circumstances, data access of specific variables will be temporarily restricted and will be shared after a data-hold period has passed. A study-specific data-hold period may be required if it is determined that release of data prior to a specific time might adversely affect ongoing or planned research.

5.1.1 Data Access Workflow



6. Data Access Applications

6.1 Eligibility

1. Data access requests must be made using the online Data Access Application form through the ACDC.
2. The Data Access Application must include a clear rationale plan for use of the data and scope (i.e. whether the proposal will provide a public benefit and which data elements are requested). The request must include a background, aims, specific objectives, a description of proposed methods, and references.
3. Users must provide approval from a Human Research Ethics Committee, or exemption, for the proposed research project.
4. Users should indicate if they seek the involvement of the Data Custodians and other parties responsible for the research from which data is requested.
5. Users must be employees or affiliates (officially part) of a recognised academic institution, health service organisation, medical research organisation, or the pharmaceutical and biotech industries from Australia and/or overseas. Users are advised to have their home organisation review the general terms and conditions for data access which are provided on the ACDC site.
6. Users must declare all perceived, potential, or actual conflicts of interest in relation to the requested data or the research that generated the requested data, including (but not limited to) its analyses and publications, and any potential commercial interests.
7. Users must declare funding sources for the research they are requesting data for.
8. Requests for data that identify individuals (such as names and contact information) will not be considered. Requests for genomic data will be considered high risk and will be handled differently.

6.2 Review and authorisation

1. Data Access Applications are initially reviewed by the Data Access Committee for completeness and compliance with the terms specified in this Policy and any additional terms that might be specific to a Data Set, and to ensure an outcome is provided to the Users in a timely manner.
2. Following Data Access Committee's review, the Data Access Application will require review and approval by the legal data owner(s), custodian(s) or authorised delegates and other parties responsible for oversight for the research from which data was requested. If the Data Access Committee finds that an application is not compliant with this policy, access to data through the ACDC will not be authorised. Data Custodians may maintain their right to grant access to data through resources other than the ACDC.
3. Data custodians will consider each request based on:
    a. Scientific merit of the proposal
    b. Publication track record of the Data Applicant.
    c. The ability of the proposed statistical analysis plan to meet the scientific objective of the research proposal
    d. Terms of participant consent under which the requested data were obtained
    e. Applicable ethical, regulatory, contractual, and legal obligations and limitations;
    f. Ethics approval (or exemption) to conduct the proposed research
    g. Potential impact of the sharing of data on ongoing or planned research, analyses, publications (including whether the proposal is already planned by the investigators);
    h. Opportunities for Data Custodians and other relevant parties to collaborate on the User's proposed research, including through an agreed *a priori* statistical analysis plan;
    i. Potential risk to the integrity or outcome of other research, business, intellectual property, interests, or reputation.
    j. Risk of participants being re-identified, or their privacy being breached
    k. Ability of the Users to demonstrate data can be kept secure.
    l. For Users who have successfully applied for other datasets: history of abiding by the conditions of any previous transfer.
4. The Data Custodians maintain the right to refuse access to data for any reason. The reason will be communicated to the Data Applicant via email if their Data Access Application is not approved.
5. The Data Access Committee and Data Custodians will aim to review completed applications within 4 weeks. If the review process is anticipated to take longer than 4 weeks the Data Applicant will be notified of the delay.
6. The Data Access Committee and Data Custodians may request further information to evaluate the application.
7. The User will then be invited to enter into a Data Access Agreement and Project Schedule if needs be.

6.3 Project Schedules

1. When a Data Access Application form describes a project that requires access to more than one Data Set, it will be necessary for the Data Applicant to enter into a Project Schedule with the ACDC.
2. The purpose of a Project Schedule is to provide an overall description of how the multiple Data Sets will be used and to consolidate the terms of the individual Data Access Agreements into a single agreement.

7. Terms of use

7.1 Data Access

Data will not be provided to the Data Applicant until all the Data Access Agreements that are required by the Data Access Application have been fully executed; and where more than one Data Access Agreement is used, a Project Schedule has also been fully executed.

7.2 Terms

Users must use the data shared with them in accordance with the approved Data Access Application and all terms and conditions outlined in the arising Data Access Agreements.
We provide here the standard terms and conditions which are common to the Data Sharing Agreements. Applicants are advised that each Data Custodian may have additional Terms and Conditions.

a. Data may not be shared with any other individual or research collaborator unless the Data Applicant submits a data access request to include other individuals as Users and this request is approved by the Data Access Committee. Additional Users must belong to an approved/registered institution and must register before accessing the ACDC.
b. Shared data will be masked in a way that prevents the identification of individuals. Users must agree not to attempt to re-identify research participants.
c. Users are not allowed to link shared data to other data from ACDC, without explicit permission from the Data Access Committee.
d. Users must provide documentation of Research Ethics Committee approval, or exemption, for the proposed research project prior to execution of the Data Access Agreement and being provided with data access.
e. ACDC Investigator Team must be appropriately acknowledged in publications and presentations arising from the research for which the requested data was used according to ICJME author and contributor guidelines[1]. See below an example of acknowledgment:
(A part of) The data used for this research was originally obtained by AAAA research project/group led by Prof. /Dr. BBBB and available at the ACDC *https://data.acdc.ozheart.org/*
f. The Data Custodian(s) must be given an opportunity to review any proposed abstracts, manuscripts or presentations that arise from research produced by the use of ACDC data. Abstracts and presentations must be forwarded to the Data Custodian(s) at least 14 days prior submission. Each manuscript to be submitted for publication must be forwarded to the Data Custodian(s) at least 30 days prior submission.
g. Users must offer the opportunity for Data Custodians to be involved in the data analysis and/or as co-authors. This allows individuals with knowledge on the original data collection to contribute to the interpretation of secondary analyses. Users must request all co-authors permission to approve any proposed publication.
h. The use of data is intended to be for research purposes. If a proposal may potentially result in a commercial outcome, then this must be specified in the Data Access Application. Further terms and conditions may apply for Users with commercial interests. These terms will be specified in the Data Access Agreement.
i. Approval to publish will be blocked where breaches to the Data Access Agreement have occurred.

[1] https://www.icmje.org/recommendations/

j. Downloaded data cannot be shared with third parties.
k. If amendments to the User's proposed project are required after the Data Access Agreement has been executed, then the Data Applicant must notify the changes to xxxx@xxx.xx. Any amendments on how the data will be used must be approved by the Data Access Committee and the Data Custodian.

Standard Terms and Conditions for Users

1. Users will have access to the requested data for the period specified in the Data Access Agreement.
2. Users must follow all directions from ACDC to ensure data security, privacy and integrity are maintained during the data sharing process and after receipt of the data.
3. To ensure we meet our ethical responsibilities and legal obligations, such as those under the *Privacy Act 1988 (Cth)* all Users who suspect or are made aware that a data breach has occurred, must immediately notify: security@biocommons.org.au to provide all available information regarding the data breach such as, but not limited to: the type of data affected; the time when the breach first occurred; the type of breach (see ACDC Data Breach Policy); etc. The Data Access Committee will then follow the relevant steps to contain, stop and, where possible, revert the data breach as mapped in the ACDC Data Breach Policy. The data breach response team will subsequently contact the Users for any further details, as required.

8. Data Access procedures

If a request for data sharing is approved, the data will be made accessible to the Users after execution of all the required Data Access Agreements and after any data-hold periods have passed. The Data Applicant will receive an email with a link to access the requested data in the ACDC.

9. Dispute resolution

In cases where a Data Access Application is rejected, the Data Applicant will be recommended to contact the Data Custodian directly.

10. Data usage auditing

Audits may be performed on a regular basis on the usage of data released to Users in order to ensure access is granted to relevant parties and used appropriately.

Additional Resources

- ACDC Data Access Agreement
- ACDC Project Schedule template
- ACDC Data Breach Policy
- ACDC Data Review and Authorisation Checklist
- ACDC Privacy Policy
- ACDC Platform - Terms of Use

References

1. Committee on Strategies for Responsible Sharing of Clinical Trial Data; Board on Health Sciences Policy; Institute of Medicine. Sharing Clinical Trial Data: Maximizing Benefits, Minimizing Risk. Washington (DC): National Academies Press (US); 2015 Apr 20. 2, Guiding Principles for Sharing Clinical Trial Data. Available from: https://www.ncbi.nlm.nih.gov/books/NBK285999/
2. Wilkinson, M. D. et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. Sci. Data 3:160018 doi: 10.1038/sdata.2016.18
3. Australian Bureau of Statistics, (ABS). Five Safes Framework. Canberra: ABS; 2021 [Available from: https://www.abs.gov.au/about/data-services/data-confidentiality-guide/five-safes-framework]
4. Australian Code for the Responsible Conduct of Research 2018. National Health and Medical Research Council, Australian Research Council and Universities Australia. Commonwealth of Australia, Canberra.
5. National Statement on Ethical Conduct in Human Research 2007 (Updated 2018). The National Health and Medical Research Council, the Australian Research Council and Universities Australia. Commonwealth of Australia, Canberra.
6. Kott, K. A., et al. (2019). Biobanking for discovery of novel cardiovascular biomarkers using imaging-quantified disease burden: protocol for the longitudinal, prospective, BioHEART-CT cohort study." BMJ Open 9(9): e028649.
7. Keech, A., et al. (2005). Effects of long-term fenofibrate therapy on cardiovascular events in 9795 people with type 2 diabetes mellitus (the FIELD study): randomised controlled trial. The Lancet 366(9500): 1849-1861.
8. Dunstan, D. W., et al. (2002). The Australian Diabetes, Obesity and Lifestyle Study (AusDiab)—methods and response rates. Diabetes Research and Clinical Practice 57(2): 119-129.

Revision History

Version	Date	Author	Reviewer	Approver	Nature of Change
0.9	14/10/2022	Dr Talia Palacios Prof Anthony Keech	ACDC Platform Governance Working Group	ACDC Platform Governance Committee	Draft version
1.0	19/03/2024	N/A	ACDC Platform Governance Working Group	ACDC Platform Governance Committee	Initial version
1.1	15/05/2025	N/A	ACDC Platform Governance Working Group	ACDC Project Management Committee	Section 2_Purpose updated Added data sharing principles and the Five Safes Framework by Dr Talia Palacios Section 5.1_ updated 6.1, 6.2 and 7.2: reworded; minor updates V1.0 reviewed and actioned all the comments and suggestions. V1.0 revised by Dr Guy Krippner keeping alignments with rest of the current project documentation. Refer track changes for all the details.